CLINICAL NEPHROLOGY & COMPREHENSIVE PATHOLOGY REFERENCE

# Kidney Disease Study Guide

Integrative analysis of renal pathologies across Allopathy, Ayurveda, Homeopathy, and Unani medical systems.

## 1. Diagnostic Biomarkers & Staging Cutoffs

Standard lab parameters utilized in clinical evaluation and diagnosis across key kidney disease categories:

| Condition | Primary Laboratory & Imaging Biomarkers | Key Diagnostic Cutoffs |
|---|---|---|
| **Chronic Kidney Disease (CKD)** | eGFR, Serum Creatinine, Urine Albumin-to-Creatinine Ratio (UACR) | eGFR < 60 mL/min/1.73m² for >3 months; UACR > 30 mg/g |
| **Nephrotic Syndrome** | 24-hr Urine Protein, Serum Albumin, Lipid Panel | Heavy Proteinuria (>3.5 g/24hr), Hypoalbuminemia (<3.0 g/dL) |
| **Glomerulonephritis** | Urine Sediment (Dysmorphic RBCs & RBC Casts), Serum C3/C4 | Active urinary sediment; Nephritic pattern with dysmorphic cells |
| **Lupus Nephritis** | Anti-dsDNA, Serum C3/C4 complement levels, Renal Biopsy | ISN/RPS Classes I–VI biopsy staging; Class IV wire-loop lesions |
| **Diabetic Nephropathy** | Urine Albumin Excretion (UACR), HbA1c, Serum Creatinine | Microalbuminuria (30–300 mg/day) progressing to >300 mg/day |
| **Acute Kidney Injury (AKI)** | Serum Creatinine, Blood Urea Nitrogen (BUN), Urine Output | KDIGO: Cr rise ≥ 0.3 mg/dL within 48h or ≥ 1.5× baseline |
| **Renal Tubular Acidosis (RTA)** | Serum Bicarbonate, Serum $K^+$, Urine pH, Urine Anion Gap | Normal Anion Gap Acidosis; Urine pH > 5.5 in Type 1 RTA |
| **Myeloma Kidney** | Serum Free Light Chains (κ/λ ratio), Urine Protein Electrophoresis | Bence-Jones proteinuria; FLC ratio > 100 with light chain casts |
| **Kidney Stones** | 24-hr Urine Chemistry (Ca, Oxalate, Uric acid), Non-contrast CT | Hypercalciuria, Hypocitraturia; CT Hounsfield Unit density |

## 2. Multi-System Therapeutic Summary

Integrative therapeutic summary across primary renal systems:

| Category | Allopathic Strategy | Ayurvedic Strategy | Homeo & Unani Strategy |
|---|---|---|---|
| **Glomerular (Filters)** | Immunosuppression (Steroids, MMF, Cyclophosphamide) + RAAS blockade | Pitta-calming herbs (Manjistha, Guduchi) & Tikta Basti detox | Apis Mellifica, Cantharis / Musaffiyat blood purifiers |
| **Renal (Vascular/Tissue)** | SGLT2 inhibitors + ACEi/ARBs for pressure relief & hemodynamic control | Punarnava, Varuna, Gokshura, & Shilajit for tissue protection | Serum Anguillae, Lycopodium / Sharbat-e-Bazoori Motadil |
| **Urological (Drainage)** | Alpha-blockers (Tamsulosin), ESWL/PCNL lithotripsy, & stenting | Pashanbhed, Kultha, & Varuna (Ashmari-hara stone expellers) | Berberis Vulgaris, Sarsaparilla / Hajrul Yahood Bhasma |